Homework 4. Trees, forests and boosting: Naming bumps on steel plates

70-467 - Machine Learning for Business Analytics

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Background. A steel plant checks the surface of its plates for faults and records where each fault sits, how large it is, and its outline and luminosity. An inspector then labels the fault. Bumps is one of seven labels.

You have 1,941 of these faults, and 402 of them are bumps. Each observation is one fault, not one plate. The file has no fault-free plates, so it cannot give a plant defect rate. Three variables describe the plate setting: the conveyor length, the steel type and the thickness. A setting is one combination of these three. It is not a plate or batch ID. The codebook on the last page gives the type of every variable.

The data are the Steel Plates Faults dataset of Buscema, Terzi and Tastle, 2010, from the UCI Machine Learning Repository, dataset 198, doi:10.24432/C5J88N, licensed CC BY 4.0. The csv is the original data with the source's variable names added as a header.

The problem. The quality team wants the inspection system to name bumps from the measurements and the plate setting, both known before an inspector labels the fault. For each fault it needs the probability that the fault is a bump. A missed bump still needs an inspector; a false bump puts the wrong label on another fault.

Inspectors can check a tree's rules by hand. The quality team must choose a model for naming bumps and say how far to trust it on a plate setting that is not in this file.

Please use the starter. Answer each question in the notebook, in the place where it asks.

The video. Imagine that you are recording a presentation for the plant's quality team. The team may use it when choosing which model names bumps and how far to trust that model on a plate setting that is not in this file. Use the answers you have already written.

Before you record, answer these for yourself. Then use what you decided in the presentation.

1. What decision will the quality team make with your model, and which model do you recommend?
2. Which variables does your model use? Why did the seven inspector flags, the A400 steel type, and any other variable you left out stay out?
3. How much better does your model rank the validation faults than the pruned tree? What do the inspectors lose without its rules?
4. On the test faults, what are your model's AUC and four counts at 0.50, next to naming no fault a bump?
5. What does the check on unseen settings say about using your model on a new plate setting? What can that check not tell you?
6. Which graph carries your recommendation? How will you order the presentation so the decision is clear before the methods?

Deliverables. Submit on Canvas by Sunday, 22 November 2026, 11:59 pm Qatar time:

- a knitted R notebook (HTML or PDF from the qmd), with the code you add and your written answers,

- at most six slides, and
- a video of at most five minutes.

The notebook questions are 45 points, the model work 30, and the slides and video 25. The notebook, the slides, and the video must use the same model and the same numbers.

The camera must be on. I need to see you delivering. Please refrain from reading the slides. That is a zero on the slides and the video. AI can assist you in preparing, but it should not be used to deliver the video. If you use AI to make the delivery for you, such as reading from a script, you will receive a zero for the video.

About AI. AI is allowed on this homework with caution, as described in the syllabus. Use it to understand the methods and to fix errors in your code, and say at the end of the notebook how you used it. The answers and the recommendation have to be yours, in your own words. If the work reads as if AI wrote the answers, I will do what I said at the start of the semester: I will ask you to come to my office and explain the whole homework to me. Depending on how that conversation goes, the grade on this homework can go as low as 0 and you could be reported for an academic integrity violation.

Please start early and give this your full effort. The homework builds on our session about tree-based methods. Lab 6 next week will help with the code for this homework. The reading is chapter 8 of *An Introduction to Statistical Learning*, second edition, pages 327 to 352. Please reach out if you need support.

Codebook. 1,941 observations, 34 variables. One observation = one fault. The source gives no engineering units or formulas, so the descriptions follow the names. Units marked unspecified are unknown. The three variables marked derived are made in the starter and are not in the csv.

Variable	Type	Unit	Description
X_Minimum, X_Maximum	numeric	unspecified	Minimum and maximum X position of the fault region, as named.
Y_Minimum, Y_Maximum	numeric	unspecified	Minimum and maximum Y position of the fault region. Origin not given.
Pixels_Areas	numeric	unspecified	Area of the fault region. The name refers to pixels.
X_Perimeter, Y_Perimeter	numeric	unspecified	Perimeter measures in X and Y, as named.
Sum_of_Luminosity	numeric	unspecified	Sum of luminosity over the fault region, as named.
Minimum_of_Luminosity	numeric	unspecified	Minimum luminosity measure, as named.
Maximum_of_Luminosity	numeric	unspecified	Maximum luminosity measure, as named.
Length_of_Conveyer	numeric	unspecified	Conveyor length. Plate setting. The source's spelling is kept.
TypeOfSteel_A300	binary	0 / 1	1 if the steel is type A300. Plate setting.
TypeOfSteel_A400	binary	0 / 1	1 if the steel is type A400. Always 1 minus A300.
Steel_Plate_Thickness	numeric	unspecified	Plate thickness. Plate setting.
Edges_Index	numeric	unspecified	Edge index. Formula not given.
Empty_Index	numeric	unspecified	Empty index. Formula not given.
Square_Index	numeric	unspecified	Square index. Formula not given.
Outside_X_Index	numeric	unspecified	Outside X index. Formula not given.
Edges_X_Index, Edges_Y_Index	numeric	unspecified	Edge indices in X and Y. Formulas not given.
Outside_Global_Index	numeric	unspecified	Global outside index. Formula not given.
LogOfAreas	numeric	unspecified	Log-area measure, as named.
Log_X_Index, Log_Y_Index	numeric	unspecified	Log indices in X and Y, as named.
Orientation_Index	numeric	unspecified	Orientation index. Formula not given.
Luminosity_Index	numeric	unspecified	Luminosity index. Formula not given.
SigmoidOfAreas	numeric	unspecified	Area-related sigmoid measure, as named.
Pastry, Z_Scratch, K_Scatch	binary	0 / 1	Inspector labels. K_Scatch is the source's spelling.
Stains, Dirtiness, Bumps	binary	0 / 1	Inspector labels. Bumps defines the outcome.
Other_Faults	binary	0 / 1	Inspector label for the remaining fault types.
y (derived)	factor	No / Yes	Yes if Bumps is 1. No if the fault has one of the other six labels.
setting (derived)	factor	group	Conveyor length, thickness and steel type combined. Used only to group faults. Not a real plate or batch.
row_id (derived)	integer	index	Order of the faults in the csv. Used only to check the split. Not a predictor.